



UNIT-5 : 5G

◆ 1. Millimeter Wave (mmWave)

What

- 5G uses high frequency (30–300 GHz)

Points

- Very high data rate
- Short range
- High path loss
- Poor penetration
- Needs line-of-sight
- Uses beamforming

Applications

- 5G hotspots
- VR/AR
- High-speed internet

✦ 1 line:

👉 “mmWave provides ultra-high speed but limited coverage”

◆ 2. Spectrum & Regulations

What

- Allocation of frequency bands for 5G

Points

- Licensed spectrum
- Unlicensed spectrum
- Sub-6 GHz + mmWave
- Managed by government bodies
- Avoids interference
- Ensures fair usage

✦ 1 line:

👉 "Spectrum management ensures efficient and interference-free communication"

◆ 3. Deployment Scenarios

Types

- Urban Macro (UMa)
- Urban Micro (UMi)
- Rural Macro (RMa)
- Indoor

Points

- Urban → high density
- Rural → large coverage
- Indoor → small cells
- Different environments need different design

✳ 1 line:

👉 “Deployment varies based on environment and user density”

◆ 4. Beamforming

What

- Directing signal toward user

Points

- Uses antenna arrays
- Improves signal strength
- Reduces interference
- Essential for mmWave
- Types:
 - Analog
 - Digital
 - Hybrid

✳ 1 line:

👉 “Beamforming focuses signal energy to improve efficiency”

◆ 5. Physical Layer Techniques

Important Techniques

- OFDM
- Massive MIMO
- Channel Coding
- Adaptive Modulation

Points

- Improves data rate
- Reduces errors
- Supports high capacity
- Handles multipath

✳ 1 line:

👉 “Physical layer ensures efficient transmission of signals”

◆ 6. Massive MIMO

What

- Large number of antennas at base station

Points

- High capacity
- Beamforming support
- Improves spectral efficiency
- Serves multiple users simultaneously
- Works with 5G NR

📌 1 line:

👉 “Massive MIMO increases capacity using many antennas”

◆ 7. Channel Models (Propagation)

What

- Mathematical models for signal behavior

Points

- Path loss
- Fading
- Delay spread
- Doppler effect
- LOS / NLOS

📌 *1 line:*

👉 “Channel models help predict signal behavior”

◆ 8. Pilot Design (Massive MIMO)

What

- Known signals sent for channel estimation

Points

- Used for CSI estimation
- Helps beamforming
- Reduces interference
- Important in massive MIMO
- Pilot contamination issue

📌 *1 line:*

👉 "Pilots are used to estimate channel conditions"

◆ 9. Resource Allocation & Transceiver Design

Points

- Allocates bandwidth & power
- Improves efficiency
- Uses scheduling algorithms
- Supports multi-user systems
- Transceiver = transmitter + receiver

📌 1 line:

👉 "Resource allocation optimizes network performance"

◆ 10. Fundamentals of Baseband & RF

Baseband

- Signal processing
- Digital domain

RF

- Wireless transmission
- Antennas

Points

- Baseband → processing
- RF → transmission
- Both required in communication

✦ 1 line:

👉 “Baseband processes signals, RF transmits signals”
